



1. GREY LEVEL TRANSFORMATION

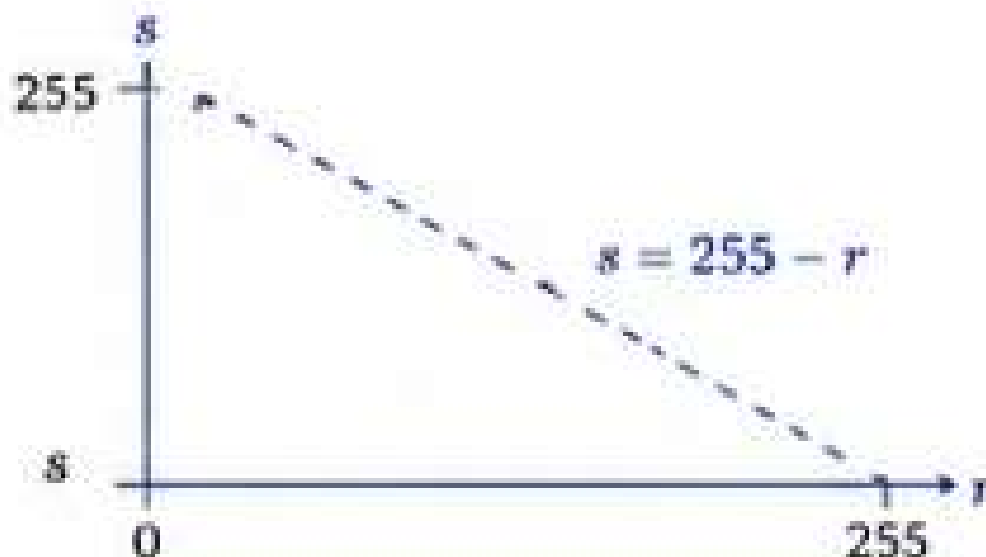
(a) Purpose

Grey level transformation is used to improve image appearance by modifying pixel intensities. It enhances contrast, stretches dynamic range, and reveals details in dark or bright regions.

(b) Negative Transformation $s = L - 1 - r$

For $L = 256$

Original (r)	0	50	100	150	200	255
Transformed (s)	255	205	155	105	55	0



Final Answer:

Transformed values = [255, 205, 155, 105, 55, 0]



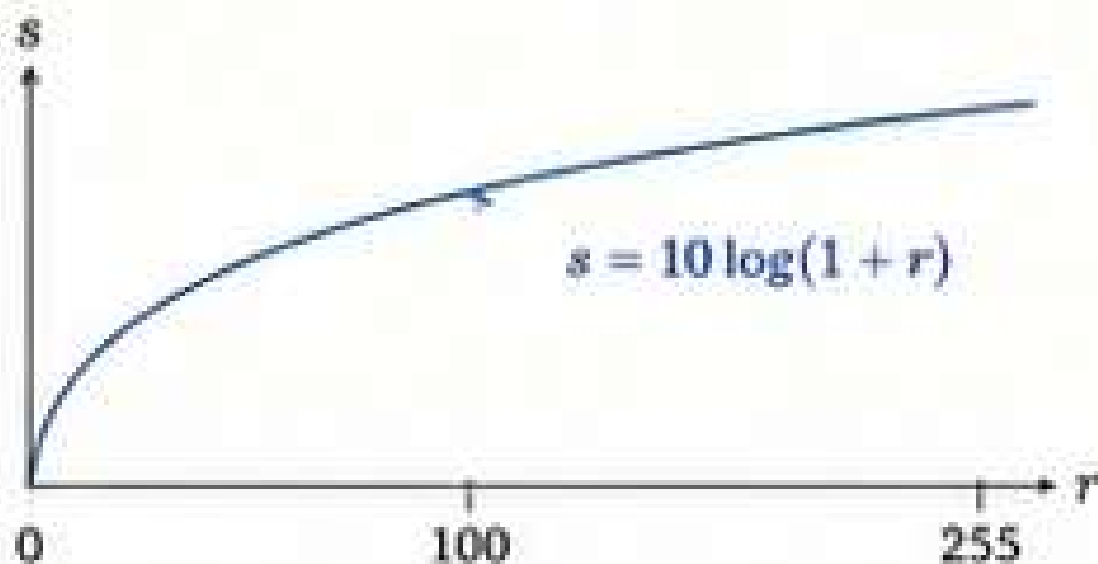
2. LOG TRANSFORMATION

(a) Explanation

Log transformation expands the range of low-intensity values while compressing high-intensity values. This enhances details in darker regions of an image.

(b) Given $s = c \log(1 + r)$, $c = 10$

r	0	10	50	100
$s = 10 \log(1 + r)$	0.00	10.41	16.99	20.04



Final Answer:

$$s = [0.00, 10.41, 16.99, 20.04]$$



3. HISTOGRAM PROCESSING

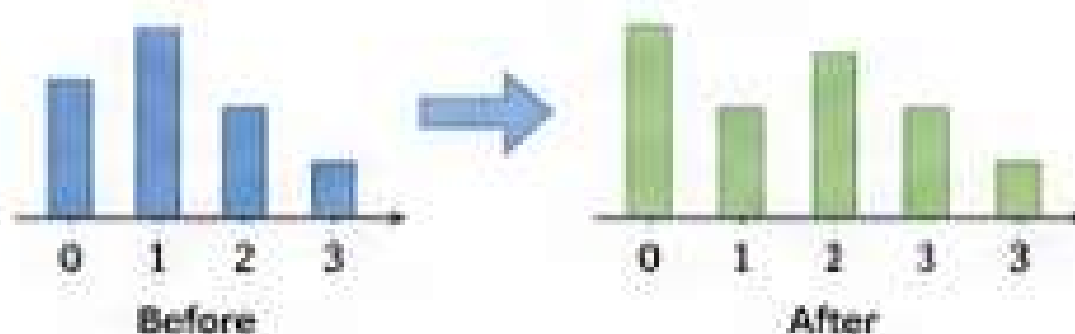
(a) Explanation

Histogram equalization redistributes pixel intensities to achieve a uniform histogram. It enhances contrast by spreading out the most frequent intensity values.

(b) 3-bit Image ($L = 8$)

Intensity (r_k)	Frequency (n_k)	Cumulative (cdf)	Equalized (s_k)
0	2	$2/16 = 0.125$	$\text{round}(7 \times 0.125) = 1$
1	2	$4/16 = 0.250$	$\text{round}(7 \times 0.250) = 2$
2	4	$8/16 = 0.500$	$\text{round}(7 \times 0.500) = 4$
3	8	$16/16 = 1.000$	$\text{round}(7 \times 1.000) = 7$

Equalization spreads intensities



Final Answer:

CDF = [0.125, 0.250, 0.500, 1.000]

Equalized Intensities = [1, 2, 4, 7]



4. SPATIAL FILTERING (SMOOTHING)

(a) Purpose

Smoothing reduces noise and fine details by averaging pixel values in a neighborhood. It improves image quality and prepares image for further processing.

(b) 3×3 Averaging Filter

$$\text{Kernel} = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Given Region:

45	50	55
40	50	60
35	45	55

$$\begin{aligned} \text{New center value} &= (45+50+55+40+50+60+35+45+55) \\ &= 435/9 = 48.33 \end{aligned}$$

Before				After (center pixel)		
45	50	55	→			
40	50	60				
35	45	55				

Final Answer:

New center pixel value = 48.33



5. SPATIAL FILTERING (SHARPENING)

(a) Explanation

Sharpening enhances edges and fine details by emphasizing rapid intensity changes. It makes image features more distinct.

(b) Laplacian Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Given Matrix:

10	10	10
10	10	10
10	10	10

$$\text{Output at center} = (4 \times 10) - (10 + 10 + 10 + 10) = 40 - 40 = 0$$

Before

10	10	10
10	10	10
10	10	14

Output (center)

0	0	0
0	0	0
0	0	0

Final Answer:

Output at center = 0



6. FREQUENCY DOMAIN FILTERING

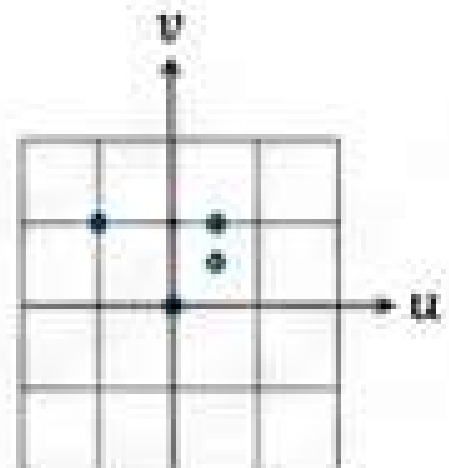
(a) Explanation

Low-pass filter (LPF) allows low-frequency components (smooth variations) and removes high-frequency components.

High-pass filter (HPF) does the opposite.

(b) Frequency Components

Component	(u, v)	Magnitude
F_1	$(0, 0)$	100
F_2	$(1, 0)$	80
F_3	$(0, 1)$	70
F_4	$(2, 0)$	30
F_5	$(1, 1)$	20



Low-pass filter retains first two (F_1, F_2)

Filtered Output:

$$\{ F_1 (0,0) = 100 , F_2 (1,0) = 80 \}$$

Final Answer:

$$\text{Filtered Output} = \{ (0,0): 100 , (1,0): 80 \}$$



7. THRESHOLDING

(a) Explanation

Global thresholding segments an image by assigning pixels to object (1) if intensity $\geq T$, otherwise to background (0).

(b) Given Pixel Values

30	65	90	120	150	200
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Threshold $T = 100$

Output (0 or 1):

0	0	0	1	1	1
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Final Answer:

Output = [0, 0, 0, 1, 1, 1]



8. EDGE DETECTION

(a) Explanation

Edge detection finds boundaries where there is a sharp change in intensity. It is essential for feature extraction.

(b) Sobel Operator (Horizontal)

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Given Matrix:

10	20	30
15	25	35
10	20	30

$$\begin{aligned} \text{Gradient at center} &= (-1 \times 15) + (0 \times 25) + (1 \times 35) \\ &+ (-2 \times 10) + (0 \times 20) + (2 \times 30) + (-1 \times 10) + (0 \times 20) + (1 \times 30) \\ &= -15 + 0 + 35 - 20 + 0 + 60 - 10 + 0 + 30 = 80 \end{aligned}$$

Before

10	20	30
15	25	35
10	20	30

G_x Output (center)

10	20	30
15	80	35
10	20	30

Final Answer:

Gradient (G_x) at center = 80



9. REGION-BASED SEGMENTATION

(a) Explanation

Region growing starts from a seed pixel and adds neighboring pixels whose intensity differs from the seed by less than a given threshold.

(b) Given Intensities

48	52	55
47	50	53
46	51	54

Seed = 50, Threshold difference = 5

Region includes pixels with $|\text{value} - 50| \leq 5$

Selected Region (1 = included, 0 = excluded)

1	1	0
1	1	0
1	1	0

Final Answer:

Region = { 48, 52, 47, 50, 53, 46, 51 }



10. COMBINED ENHANCEMENT & SEGMENTATION

(a) Explanation

Smoothing reduces noise and small variations that may cause incorrect segmentation. It improves the accuracy of thresholding.

(b) Given Image Values

45	47	50	52	48
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Step 1: Averaging (mean = 48)

Smoothed Values				
47	47	48	49	48

Step 2: Threshold at $T = 48$ ($\geq 48 \rightarrow 1$, else 0)

Final Segmented Output				
0	0	1	1	1

Final Answer:

Final Output = [0, 0, 1, 1, 1]